

at the center of another, passing through the centers of the cells on its way horizontally, vertically, or turning 90 degrees. Two arrows cannot occupy the same cell.

QUESTION. Is there at least one solution to the puzzle, i.e., a mapping from the set of white cells to the set $\{\emptyset, \square\}$ that satisfies the rules of the puzzle (each block contains exactly one square placed on the arrow, each arrow passes through at least two blocks, the sizes of the blocks increase in the direction of the arrow, maintaining the shape)?

We also consider a problem of COUNTING EVOLOMINO, which has the same instance but asks how many distinct solutions the puzzle has.

3 Evolomino $\in$ NP

Lemma 1. *EVOLOMINO $\in$ NP.*

Proof. EVOLOMINO is a decision problem, which means that to prove that it belongs to the class NP, it is sufficient to show that verifying that a given solution satisfies all the rules of the puzzle is performed in polynomial time.

Let us be given an EVOLOMINO puzzle on a rectangular board of size $p \times q$. If we guess some solution to the puzzle, then checking its correctness will require:

- Rule: “Each block must contain exactly one square placed on a pre-drawn arrow.”

Since each board cell can belong to no more than one block, we have at most $O(pq)$ blocks. The complexity of traversing one block is linear in its size, which does not exceed the size of $p \times q$ of the entire board. Thus, the total complexity of verifying this rule is $O(p^2q^2)$.

- Rule: “Each arrow must pass through at least two blocks”.

Checking this rule requires a single pass through each pre-drawn arrow. Since the total length of all arrows does not exceed the size of the entire board, the complexity of verifying the rule is $O(pq)$.

- Rule: “The second and later blocks on the route of an arrow from start to finish must progress by adding one square to the previous block without rotating or flipping”.

Let us consider two consecutive blocks in the direction of the arrow, both of size no greater than $O(pq)$. Let’s try to exclude each square of the second block one by one, after which it remains to go through both blocks and check that they coincide. Therefore, the complexity of verifying the rule for two consecutive blocks does not exceed $O(p^2q^2)$.

It remains to note that the number of pairs of consecutive blocks does not exceed the total length of all pre-drawn arrows, i.e., $O(pq)$. Which makes the overall complexity of verifying the rule equal to $O(p^3q^3)$.

Summing up the complexity of all the rules, we find that a given puzzle solution can be verified in polynomial time $O(p^3q^3)$. Note that this is a fairly rough estimate, but to prove that EVOLOMINO $\in$ NP, it is sufficient to be polynomial in the size of the board $p \times q$.